

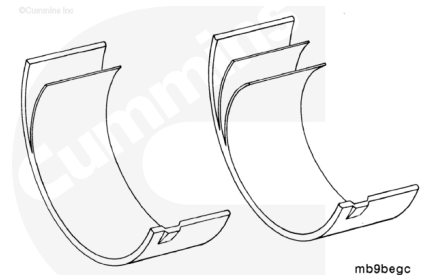
Trainings ▾

General Information

In general, various types of connecting rod bearings have been used in B series engines:

- Bi-Metal overlays
- Tri-Metal overlays
- Narrowed bearings (for some naturally aspirated engines).

Make sure to reference the appropriate part information resources when replacing the connecting rod bearings to ensure the correct connecting rod bearings are installed.

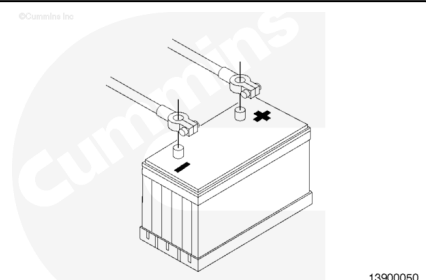


Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged

contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

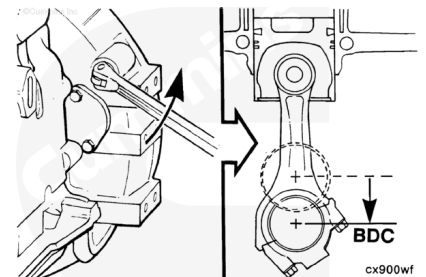
- Drain the lubricating oil. Refer to Refer to Procedure Procedure 007-037
(/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)
- Remove the oil pan and oil pan gasket. Refer to Refer to Procedure Procedure 007-025
(/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)
- Remove the lubricating oil suction tube. Refer to Refer to Procedure Procedure 007-035
(/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)
- If equipped, remove the block stiffener plate. Refer to Refer to Procedure Procedure 001-089
(/qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html).

Remove

Note : The cylinder head does not need to be removed if the connecting rod bearings are being inspected or replaced.

⚠ CAUTION ⚠

If equipped with fracture split connecting rods, be careful not to damage the fractured split surface on the connecting rod or connecting rod cap while the connecting rod cap is removed. If the fractured split surface is damaged, the connecting rod and connecting rod cap must be replaced to help reduce the possibility of engine damage.



Using the barring tool, Part Number 3824591, bar the engine over to bottom dead center of the connecting rod bearing to be removed.

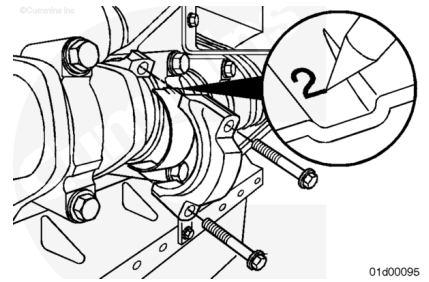
Note : Mark each connecting rod and cap to identify cylinder location.

Service Tip: Remove and install only one connecting rod cap at a time to avoid mismatching connecting rods with the wrong connecting rod cap.

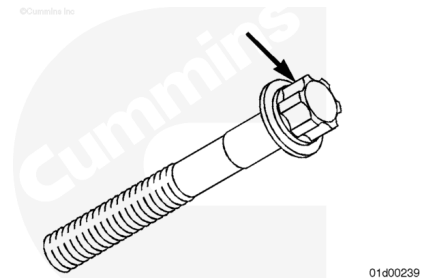
Remove the connecting rod bolts and connecting rod cap.

Remove the lower end bearing.

Mark the cylinder number and the letter "L" (lower) on the flat surface of the bearing tang.

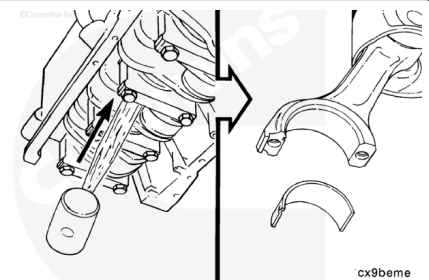


Some engines are equipped with an external Torx head.



⚠ CAUTION ⚠

Do not damage the J-jet piston cooling nozzles when sliding the connecting rod into the cylinder away from the crankshaft. Engine damage can result.



Note : Slight bending of the j-jet piston cooling nozzles can result in severe engine damage. Replace piston cooling nozzles if any are bent or damaged during assembly or disassembly.

If equipped, remove the J-jet piston cooling nozzle for each cylinder that the connecting rod bearings will be removed.

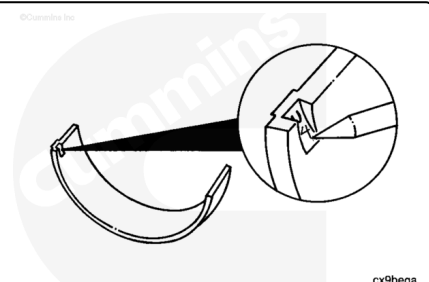
Refer to Procedure 001-046

(/qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html).

Push the connecting rod into the cylinder, away from the crankshaft, to allow the upper bearing shell to be removed.

Mark the connecting rod bearings for position and cylinder number as they are removed.

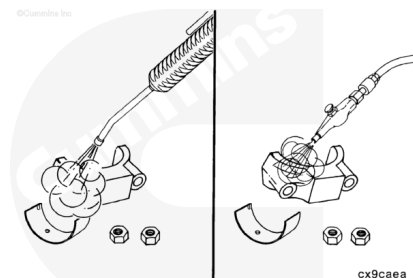
- Example: U1 - upper rod bearing for cylinder number 1.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Wash the bearing and connecting rod caps.

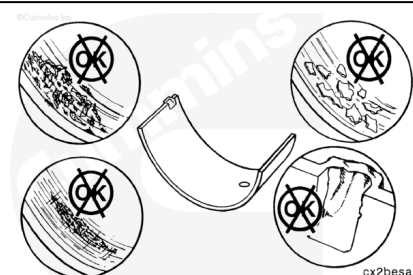
Dry with compressed air.

Inspect the connecting rod caps, connecting rod bearing saddles and capscrews for nicks, cracks, burrs, scratches, or fretting.

Inspect the bearings for damage.

Replace any bearings with the following damage:

- Pitting
- Flaking
- Corrosion
- Lock tang damage
- Scratches.

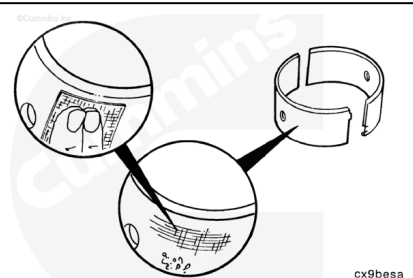


Inspect the bearing shell seating surface for nicks or burrs.

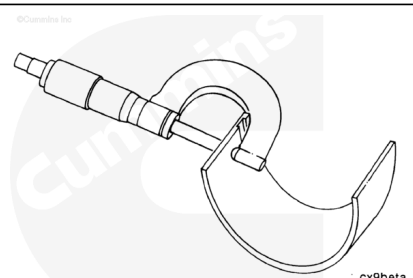
If nicks or burrs can **not** be removed with an abrasive pad, Part Number 3823258 or equivalent, the bearings **must** be replaced.

Note : If bearings are damaged they **must** be replaced as a set.

Note : For more detailed information of bearing damage, see "Analysis and Prevention of Bearing Failures" Bulletin No. 3810387.



Measure the rod bearing shell thickness with an outside micrometer that has a ball tip.



Standard Connecting Rod Bearing Dimensions B3.9, B4.5 RGT, and B5.9 Engines

	mm		in
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Standard Connecting Rod Bearing Dimensions B3.9, B4.5 RGT, and B5.9 Engines

	1.955	MIN	0.0770
	1.968	MAX	0.0775

Oversize Connecting Rod Bearing Dimensions B3.9, B4.5 RGT, and B5.9 Engines

	mm		in
0.25 mm [0.010 in]	2.080	MIN	0.0819
	2.093	MAX	0.0824
0.50 mm [0.020 in]	2.205	MIN	0.0868
	2.218	MAX	0.0873
0.75 mm [0.030 in]	2.330	MIN	0.0917
	2.343	MAX	0.0922
1.00 mm [0.040 in]	2.455	MIN	0.0967
	2.468	MAX	0.0972

Standard Connecting Rod Bearing Dimensions B4.5 Engines

	mm		in
	1.809	MIN	0.0712
	1.818	MAX	0.0716

Oversize Connecting Rod Bearing Dimensions B4.5 Engines

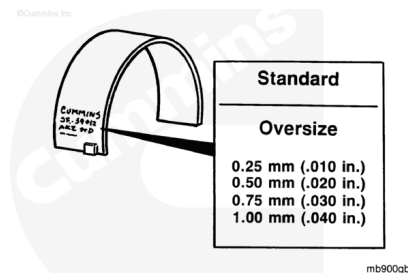
	mm		in
0.25 mm [0.010 in]	1.934	MIN	0.0761
	1.943	MAX	0.0765
0.50 mm [0.020 in]	2.059	MIN	0.0811
	2.068	MAX	0.0814
0.75 mm [0.030 in]	2.184	MIN	0.0860
	2.193	MAX	0.0863
1.00 mm [0.040 in]	2.309	MIN	0.0909
	2.318	MAX	0.0913

Discard a bearing shell if its thickness is below the minimum specification.

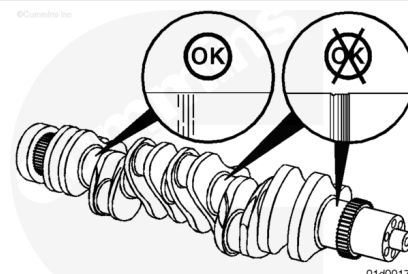
Note : Connecting rod bearings are identified with a part number and size stamped on the back side.

Determine the size of the removed rod bearing and obtain a set of the same size.

Oversize service rod bearings are available for use with crankshafts that have been machined undersized. See appropriate parts catalog.



Check the rod bearing journals for damage or excessive wear. Minor scratches are acceptable.



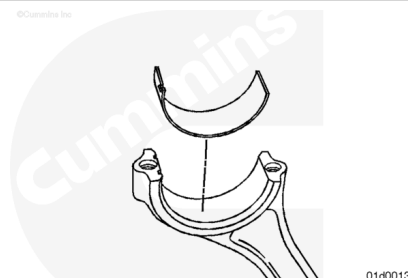
Install

Note : Used connecting rod bearings must be installed in their original locations.

Note : To help reduce the possibility of engine damage, make sure that the backside of the bearing, and connecting rod surface that the bearing contacts is clean and dry.

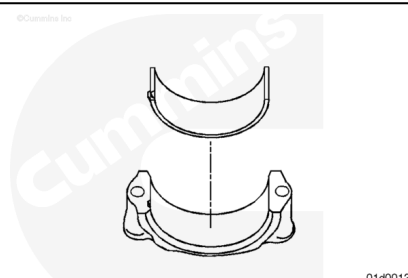
Install the upper bearings in the connecting rod. Make sure that the tang is aligned.

Note : Some B3.9 naturally aspirated engines use an upper connecting rod bearing narrower than the connecting rod mounting surface. The narrower bearing must only be installed in the upper location.



Install the lower bearings in the connecting rod caps. Make sure the tang is aligned.

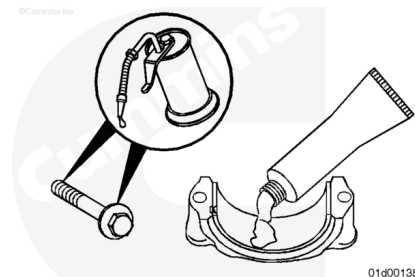
Note : For engines equipped with fracture split connecting rods, the upper and lower connecting rod bearing are not interchangeable.



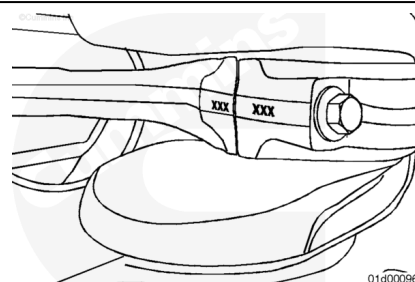
Note : For engines equipped with fracture split connecting rods, the fracture split surface on the connecting rod and connecting rod cap **must** be kept dry and clean to ensure proper mating of the two surfaces.

Lubricate the bearing surfaces with Assembly Lube, Part Number 3163087 or equivalent.

Lubricate the connecting rod bolt threads and under the heads with clean 15W-40 oil.



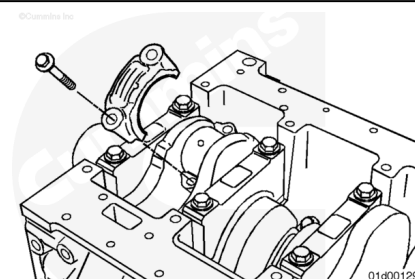
When installing the connecting rod cap, the numbers stamped on the connecting rod cap and connecting rod **must** match and be on the same side of the connecting rod, to prevent engine damage during engine operation.



Install the connecting rod caps on the corresponding connecting rod.

Install and hand-tighten the connecting rod capscrews.

Note : The long end of the connecting rod **must** be on the **exhaust** side of the engine.



Tighten the connecting rod capscrews.

B3.9, B4.5 RGT and B5.9 engines

Torque Value:

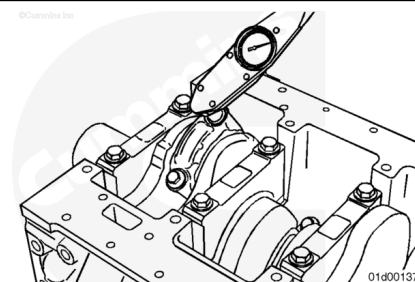
1. 30 $\text{n}\cdot\text{m}$ [22 ft-lb]
2. 60 $\text{n}\cdot\text{m}$ [44 ft-lb]
3. Rotate 60 degrees.

B4.5 engines

Torque Value:

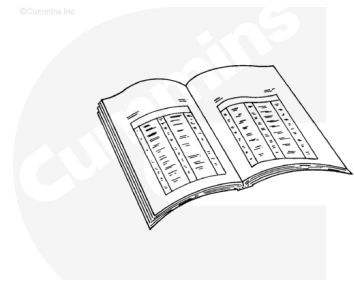
1. 25 $\text{n}\cdot\text{m}$ [18 ft-lb]
2. Rotate 60 degrees.

Note : After tightening the capscrew for each connecting rod, rotate the crankshaft 360 degrees to make sure the crankshaft rotates freely.



Finishing Steps

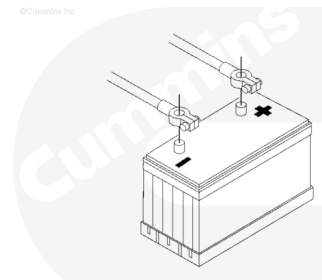
- If equipped and previously removed, install the J-jet piston cooling nozzles. Refer to Procedure 001-046 (</qs3/pubsys2/xml/en/procedures/40/40-001-046-tr.html>)
- If equipped, install the block stiffener plate. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/40/40-001-089-tr.html>)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>)
- Install the oil pan and oil pan gasket. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)



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⚠ WARNING ⚠

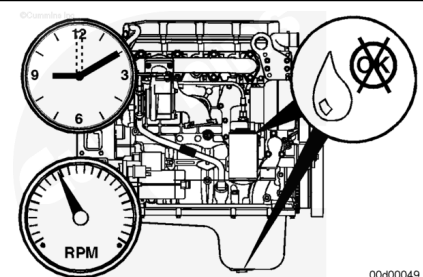
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- Connect the batteries.

- Operate the engine and check for leaks and proper oil pressure.



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